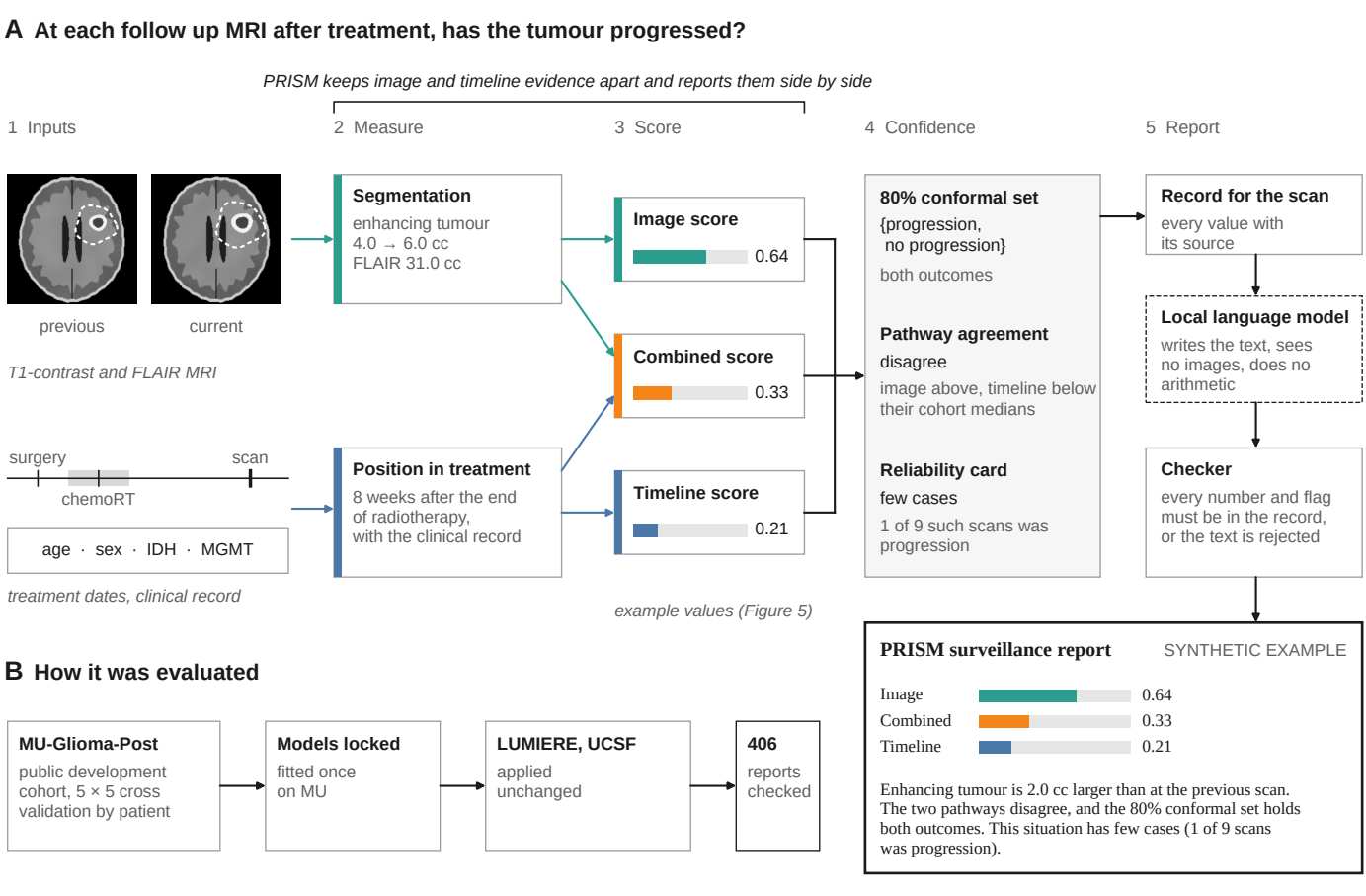

A At each follow up MRI after treatment, has the tumour progressed?
PRISM keeps image and timeline evidence apart and reports them side by side
1 Inputs
2 Measure
3 Score
4 Confidence
5 Report
previous
current
T1-contrast and FLAIR MRI
surgery
scan
chemoRT
age · sex · IDH · MGMT
treatment dates, clinical record
Segmentation
enhancing tumour
4.0 → 6.0 cc
FLAIR 31.0 cc
Position in treatment
8 weeks after the end
of radiotherapy,
with the clinical record
Image score
0.64
Combined score
0.33
Timeline score
0.21
example values (Figure 5)
80% conformal set
{progression,
no progression}
both outcomes
Pathway agreement
disagree
image above, timeline below
their cohort medians
Reliability card
few cases
1 of 9 such scans was
progression
Record for the scan
every value with
its source
Local language model
writes the text, sees
no images, does no
arithmetic
Checker
every number and flag
must be in the record,
or the text is rejected
PRISM surveillance report
SYNTHETIC EXAMPLE
Image 0.64
Combined 0.33
Timeline 0.21
Enhancing tumour is 2.0 cc larger than at the previous scan.
The two pathways disagree, and the 80% conformal set holds
both outcomes. This situation has few cases (1 of 9 scans
was progression).
B How it was evaluated
MU-Glioma-Post
public development
cohort, 5 × 5 cross
validation by patient
Models locked
fitted once
on MU
LUMIERE, UCSF
applied
unchanged
406
reports
checked